

Classical Mechanics - Home Assignment 4

Hamiltonian Dynamics and Canonical Transformations (November 2024)

1. One gets Hamiltonian $H(q, p, t)$ by replacing *all* generalised velocities $\dot{q}_i$ by the corresponding momenta. Find the function $R(q_i, p_j, \dot{q}_k, t)$ with $i = 1, 2, \dots, f$, $j = 1, 2, \dots, s$, $k = s + 1, s + 2, \dots, f$ obtained by replacing *only* s velocities by the corresponding momenta and keeping others unchanged (where f is the number of degrees of freedom). Also, write down the Hamilton's equations of motion when the coordinates $q_1, q_2, \dots, q_s$ are cyclic. (R is called the Routhian.)
2. The Lagrangian for a system of one degree of freedom can be written as

$$L = \frac{1}{2}m [\dot{q}^2 \sin^2(\omega t) + q\dot{q}\omega \sin(2\omega t) + q^2\omega^2] .$$

What is the corresponding Hamiltonian? Is it conserved?

Introduce a new coordinate $Q = q \sin(\omega t)$ and find the Lagrangian in terms of the new coordinate and the corresponding Hamiltonian. Is it conserved ?

3. In each of the following cases, show that the transformation is canonical and find the generating function :
 - (a) $Q = \log(1 + \sqrt{q} \cos p)$, $P = 2(1 + \sqrt{q} \cos p) \sqrt{q} \sin p$
 - (b) $Q = p + iaq$ $P = (p - iaq)/(2ia)$ where $i = \sqrt{-1}$ and $a = \text{constant}$
 - (c) $Q = \log(\frac{1}{q} \sin p)$ $P = q \cot p$
 - (d) $Q = \tan^{-1} \frac{\alpha q}{p}$, $P = \frac{\alpha q^2}{2} \left(1 + \frac{p^2}{\alpha^2 q^2}\right)$
4. If two canonical transformations $(q, p) \rightarrow (Q, P)$ and $(Q, P) \rightarrow (Q', P')$ are generated by $F_2(q, P)$ and $G_2(Q, P')$ respectively, then show that the transformation $(q, p) \rightarrow (Q', P')$ is a canonical transformation generated by

$$H_2(q, P') = F_2(q, P) + G_2(Q, P') - \sum_i Q_i P_i$$

5. For a canonical transformation $(q, p) \rightarrow (Q, P)$, let $J = \partial(Q, P)/\partial(q, p)$ be the Jacobian matrix. Also let

$$M = \begin{pmatrix} \underline{0} & \underline{1} \\ -\underline{1} & \underline{0} \end{pmatrix}$$

be a $2f \times 2f$ matrix where $\underline{1}$ and $\underline{0}$ are $f \times f$ unit matrix and null matrix respectively. Prove that

$$J^T M J = M$$

(This is called symplectic transformation. Hence, canonical transformation is symplectic.)